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Late Cretaceous paleomagnetic results from Northeast Asian continental margin: The Sikhote Alin mountain range, eastern Russia

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Abstract. Welded tuffs in the Upper Cretaceous Kisin Group have been collected for paleomagnetic study from 24 sites in Sikhote Alin, eastern Russia. Normal paleomagnetic directions from 11 sites are regarded as primary directions, whereas reversed paleomagnetic directions from 13 sites are ascribed to secondary origin. The normal directions of the Kisin Group together with the characteristic directions of the Sijanov Group yield the Late Cretaceous characteristic direction of Sikhote Alin (D=355.0°, I=59.8° with α_{95}=7.2°). A significant separation (15.4°±11.9°) of pole positions is observed between Sikhote Alin (85.9°N, 11.3°E with A_{95}=10.1°) and the North China Block. Sikhote Alin has been subjected to counterclockwise rotation through 20.5°±17.7° with respect to North China Block. The rotation occurred during 66-51 Ma, and is ascribed to sinistral motion along the central Sikhote Alin fault.

Introduction

Large tectonic events have occurred in zones between oceanic and continental plates such as northeastern Pacific-North America and southeastern Pacific-South America. A typical continental margin exists between the Asian continent and the Pacific and the Philippine Sea plates from the Kamchatka peninsula to the Philippines. Subduction-related tectonic phenomena have occurred in this continental margin since Mesozoic time. Widespread Mesozoic accretionary complexes developed associated with subduction of oceanic plates (Sengör and Natal'in, 1996). The oceanic plate motion activated Mesozoic NE-SW trending sinistral fault system (Xu et al., 1987). Westward suturing of microcontinents occurred in Mesozoic and Cenozoic times (Faure and Natal'in, 1995). The opening of back arc basins in Tertiary gave rise to the disruption of geological structures in the continental margin.

Sikhote Alin is a fold belt situated to the east of the major cratons of Northeast Asia, facing the Japan Sea to the east (Fig. 1a). Prior to the opening of the Japan Sea, it was a continental margin of Northeast Asia. In Late Cretaceous to Paleogene time Sikhote Alin suffered a tectono-thermal event. The tectono-thermal event has associated with extensive igneous activity and NE-SW trending fault motion along the central Sikhote Alin fault evidenced by NE-SW stretching lineation in metamorphic rocks (Faure et al., 1995). Sikhote Alin is a suitable area to study the tectonics of the continental margin since the Late Cretaceous. We present Late Cretaceous paleomagnetic results from Sikhote Alin to investigate the tectonic evolution of the marginal area since the Late Cretaceous.

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Geological setting and sampling

The basement of the Sikhote Alin mountain range consists mainly of a Late Jurassic accretionary complex and Early Cretaceous turbidites. The basement is cut by a large sinistral fault trending NE-SW (central Sikhote Alin fault). On the basis of displacement of the Amur suture (Natal'in, 1993), the fault has a lateral offset of about 200-250 km. In the Late Cretaceous to Paleogene, igneous activity occurred in the eastern Sikhote Alin, resulting in large scale volcanism (eastern Sikhote Alin volcano-plutonic belt). The Late Cretaceous to Paleogene volcanic rocks have been divided into six volcanic groups, in ascending order (Matunin, 1988): the Sinancha, Kisin, Sijanov, Bogopol, Tadusi, and Suvolov Groups. The Kisin Group contains elements of the subtropical Monastyrsky horizon flora which is indicative of the Late Campanian (Krassilov, 1990). The Sijanov and Bogopol welded tuffs are K-Ar dated at 66.0 Ma and 50.6 to 52.6 Ma, respectively (Otofuji et al., 1995).

Welded tuffs of the Kisin Group were collected at 24 sites in the southern part of Sikhote Alin for paleomagnetic measurements (Fig. 1b). The eutaxitic structure of each site was measured as a bedding plane for tilt correction. The tilt of bedding planes is gentle for all the sites (< 35°). The tilting of the Kisin Group occurred prior to Neogene because no tilting is observed in Neogene rocks in this area.

Paleomagnetism

Progressive thermal demagnetization removed a low temperature component between 250°C and 560°C. In-situ directions of the low temperature component for nine sites (SA07, 21, 48, 57, 63, 70, 71, 72 and 73) are close to that of the present field (D=349.9°, I=59.3°), whereas a southerly declination with negative inclination were observed from sites SA81 and 85 (Fig. 2a-c). The remaining thirteen sites provide inconsistent directions (α_{95}>20°) of this component. The high temperature component decays straightforwardly to the origin. Since the unblocking temperatures range between 590°C and 680°C (Fig. 2d), the high temperature component resides both magnetite and hematite. The high temperature component is recognized as a characteristic magnetization component for a sample, which is identified from all the 24 sites (Table 1). Eleven sites show normal polarity and the remaining 13 sites show reversed polarity.

The characteristic directions from 11 sites with normal polarity cluster after tilt correction: the precision parameter increases from

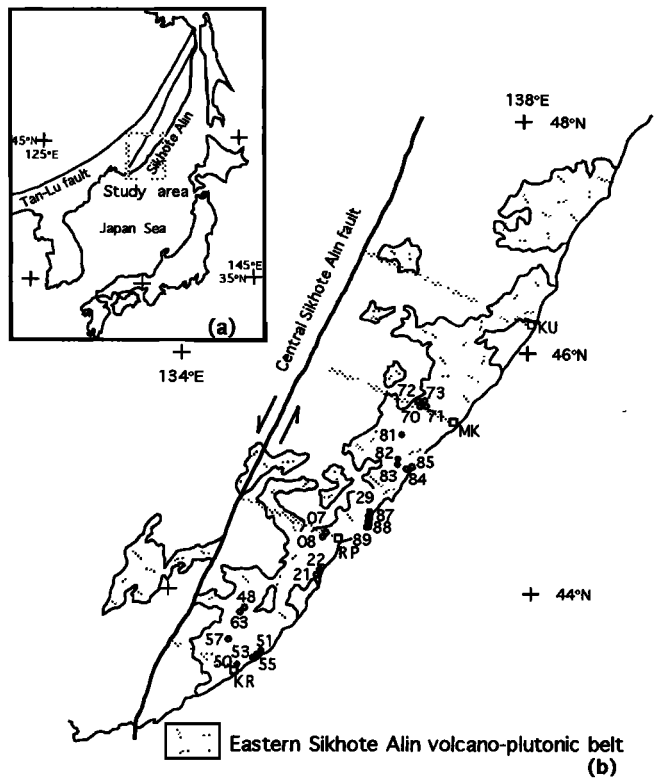


Fig. 1. (a) Sketch map of Sikhote Alin with adjacent area. (b) Geological map of the study area. Solid (open) circles denote the paleomagnetic sampling sites of the Kisin Group which yield normal (reversed) paleomagnetic directions. Stippled lines are major branch faults of the central Sikhote Alin fault. KU = Kuzunetsovo; MK = Malaya Kema; RP = Rudnaya Pristan; KR = Kamen Ribolov.

8.6 to 19.5 (Fig. 3a). The fold test is positive at the 95% confidence level (McFadden, 1990), indicating that the remanent magnetization is of prefolding origin. A mean direction from the 11 sites after tilt correction is D=353.3°, I=59.4° with α_{95}=10.6°.

The reversed directions divide into three distinct groups before tilt correction (Fig. 3b): a group of six sites with southerly declination and intermediate inclination, five sites with southeasterly declination and moderate inclination, and two sites with northerly declination and steep inclination. The grouping

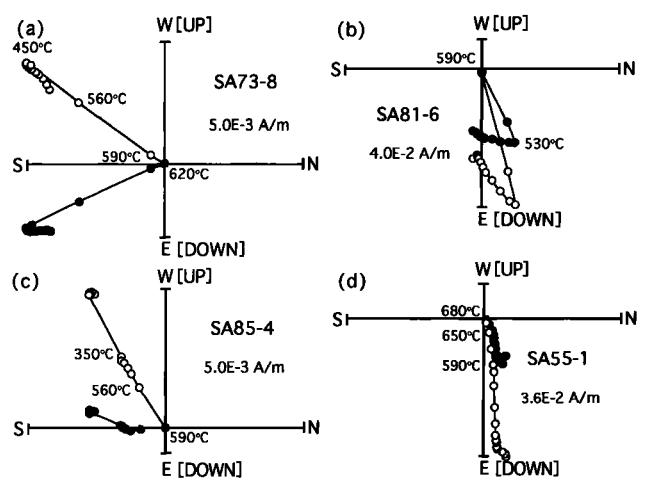


Fig. 2. Orthogonal plots of thermal demagnetization of the Kisin Group (before tilt correction). Solid and open symbols represent projection on horizontal and vertical planes, respectively.

Table 1. Paleomagnetic results of the Kisin Group.

Site	N	In Situ D(°)	In Situ I(°)	Tilt Corrected D(°)	Tilt Corrected I(°)	k	α_{95}(°)	Locality Lat(° N)	Locality Long(° E)	Strike/Dip (°)	(°)
SA29	10	328 2	61 3	329 4	49 4	382 4	2 5	44 63	136.20	243	12
SA50	10	336.9	54 6	348.5	36 7	362 1	2 5	43 36	134 78	285	21
SA51	8	58 6	66.4	5 0	69 3	45 9	8 3	43 47	134 97	205	21
SA53	10	16 6	85 2	358.2	71 5	116 7	4 5	43 45	134 94	262	14
SA55	10	46.8	73.1	352 1	62 7	88 1	5.2	43 46	134 96	223	23
SA81	10	52 0	56.9	44.0	63 5	122 1	4.4	45 31	136.57	173	8
SA82	8	26 4	42.6	21 0	64 3	450.6	2 6	45.11	136 53	124	22
SA83	7	31.1	50 5	31.1	50.5	63.5	7 6	45.10	136.53	0	0
SA87	10	319.0	62 1	343 0	48 5	550 1	1 9	44 61	136 22	292	21
SA88	9	300 1	35 1	308.7	63 3	56 5	6.9	44 60	136.22	20	29
SA89	9	309 5	34 7	330 0	45 2	385 6	2 6	44 60	136 22	343	26
Normal	11	354 0	64 0			8 6	16 5	44 00	135 00		
Mean				353 3	59 4	19.5	10 6				
SA07	9	185 4	-65 1	238 5	-84.2	101.5	5 1	44 47	135 70	83	22
SA08	6	183 3	-59 4	169 7	-61 5	127 9	5 9	44 47	135 70	80	15
SA48	9	194 0	-67 8	149 0	-57.8	27 7	10.0	43 81	134 83	194	24
SA57	7	185 8	-64 3	199 8	-37.3	215 2	4.1	43.57	134 66	305	29
SA63	10	202 1	-52 1	197 0	-53.0	516.0	2.1	43 79	134 78	187	4
SA85	8	170 6	-58 1	224 3	-60 7	190 1	4 0	45.03	136 64	22	30
Reversed	6	186.9	-61 4			102	6 7	44 00	135.00		
Mean				191 1	-61 4	16 0	17 3				
SA70	10	117.5	-40.8	121 5	-34 7	677.0	1 9	45 62	136 82	250	8
SA71	10	127.0	-42 7	123 0	-40 6	487 3	2 2	45 62	136 84	150	5
SA72	10	118 6	-49 1	127 4	-39.7	902 7	1 6	45 63	136 81	257	13
SA73	10	126 9	-36 7	136.3	-26 6	55 0	6 6	45.63	136 81	278	18
SA84	9	119.0	-39 2	130 6	-72 6	378.6	2 6	45 03	136 64	22	34
Reversed	5	121 5	-42 1			189 8	5 6	44 00	135.00		
Mean				127 3	-42 8	20 2	17 5				
SA21	10	322.6	-79 1	235 7	-88 2	120 0	4 4	44 15	135.63	242	11
SA22	10	26 5	-84 9	104 1	-61 4	124 0	4 4	44 16	135 65	205	28
Reversed	2	339 4	-83 0			137 3	——	44 00	135 00		
Mean				138 8	-81 5	40 5	——				
All Sites	24	340.3	64.2			9 3	10 3	44.00	135.00		
Mean	24			342.0	60 7	11 4	9 2				

Mean pole	N	In Situ Lat(° N)	In Situ Long(° E)	Tilt Corrected Lat(° N)	Tilt Corrected Long(° E)	A_{95}(°)
Normal	11	85.1	113 0			23 2
				85.2	19.6	14 3
All Sites	24	77 8	76 6			14 8
				78 4	53.9	12 4
Sikhote Alin (K2)	22			85 9	11 3	10.1

N = the number of the samples in calculation for the mean directions; D and I = declination and inclination, respectively; k = the precision parameter; α_{95} and A_{95} = radius of the cone of 95% confidence; Lat = north latitude; Long = east longitude.

disappears after tilt correction. The tilt correction of all the 13 sites reduces the precision parameter from 9.9 to 8.6, but this is not significant.

We perform the fold test and the reversal test for all 24 paleomagnetic directions of the Kisin Group. Although the reversal test for the 11 normal and 13 reversed directions is positive at the 95% confidence level (McFadden and McElhinny, 1990), the fold test for all the 24 directions is inconclusive at the 95% confidence level (McFadden, 1990). This suggests that the reversed directions are responsible for the inconclusive fold test of the Kisin Group.

The inconclusive fold test for the reversed directions can be ascribed to remagnetization. The tilt correction for each group of reversed polarity shows a postfolding nature. The precision parameter decreases significantly after the tilt correction for each group (kg/ks = 6.4, 7.6 and 9.2, respectively). The first two groups yield negative fold tests at the 95% confidence level (McFadden, 1990), and the two directions in the third group diverge after tilt correction.

The in-situ directions of the first group of reversed polarity (D=186.9°, I=-61.4° with α_{95}=6.6°) are close to the tilt-corrected directions of the Sijanov Group of 66 Ma (D=177.1°, I=-59.6°

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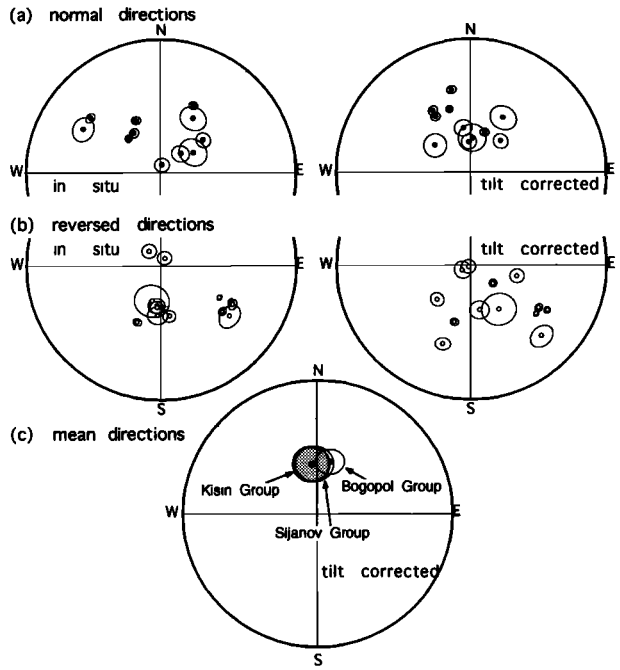


Fig. 3. Equal area projections of the site mean paleomagnetic directions (with associated 95% confidence circles) of the Kisin Group: (a) normal directions; (b) reversed directions. (c) Tilt corrected mean directions (with associated 95% confidence circles) of the Kisin (Late Cretaceous), Sijanov (66 Ma) and Bogopol (52 Ma) Groups. Directions of the Sijanov and Bogopol Groups are after Otofuji et al. (1995).

with α_{95}=11.2°) and the Bogopol Group of 51 Ma (D=193.7°, I=-57.6° with α_{95}=8.1°) (Otofuji et al., 1995). According to the geological map of Bazhanov and Oleinik (1986), the sampling localities of the Kisin Group with reversed polarity are located within 2-5 km of the distribution area of the Sijanov and Bogopol Groups with 500-1300 m thickness and coeval granitic intrusions. The remagnetization is probably due to igneous activity at some stage between 66 and 51 Ma.

The directions of the second and third groups of reversed polarity (D=121.5°, I=-42.1° and D=339.4°, I=-83.0°, respectively) are observed as paleomagnetic directions during 66-51 Ma in Sikhote Alin. In-situ directions of the second group are parallel to three tilt-corrected directions of the Sijanov Group (sites SA11, 23 and 67, in Otofuji et al., 1995), while in-situ directions of the third group are parallel to three tilt-corrected directions of the Bogopol Group (sites SA46, 59 and 64, in Otofuji et al., 1995). The reversed directions of the Kisin Group can be ascribed to remagnetized directions of geomagnetic field during 66-51 Ma.

Variation of paleomagnetic directions among the three reversed groups may be attributed to secular variation of the geomagnetic field. Different paleomagnetic directions imply close connection between geomagnetic behavior and remagnetization: sporadic thermal activity formed the Sijanov and Bogopol Groups in Sikhote Alin. The igneous activity occurred in different periods, so that different geomagnetic field directions are recorded in the remagnetized rocks in the Kisin Group.

It should be noted that there may be disagreement between a true paleohorizontal plane and an observed plane expected from eutaxitic structure. The bedding planes expected from eutaxitic structure are, however, roughly consistent among neighboring sites in an area (e.g., sites SA70, 71, 72 and 73). This suggests that the bedding planes indicate the paleohorizontal plane in such area although the expected paleohorizon undulates somewhat.

Discussion

The mean direction of normal polarity of the Kisin Group (D=353.3°, I=59.4° with α_{95}=10.6°) is regarded as the characteristic direction for Sikhote Alin of the Late Cretaceous time. The characteristic directions of prefolding origin are obtained from wide area covering 200 × 150 km in latitudinal and longitudinal directions. This normal direction is free from remagnetization due to the thermal event which formed the Sijanov and Bogopol Groups, because the Sijanov and Bogopol Groups show only reversed characteristic directions except for one site (Otofuji et al., 1995). The characteristic direction of the Kisin Group is presumably of primary origin.

The Late Cretaceous to early Tertiary paleomagnetic directions for Sikhote Alin have been established by directions from the Kisin, Sijanov and Bogopol Groups (Fig. 3c) (Otofuji et al., 1995). Although these three directions are indistinguishable within the 95% confidence limit, it is remarkable that the characteristic directions of the Kisin and Sijanov Groups are parallel. We calculate a Late Cretaceous mean direction (D=355.0°, I=59.8° with α_{95}=7.2°) and a paleomagnetic pole (85.9°N, 11.3°E with A_{95}=10.1°) for Sikhote Alin on the basis of the directions from the Kisin and Sijanov Groups. Because the Kisin and Sijanov Groups cover a fairly long duration between Campanian to 66 Ma, the mean direction is regarded as a representative of the Late Cretaceous Sikhote Alin.

The Late Cretaceous pole for Sikhote Alin is compared with the Cretaceous one for North China Block of 78.6°N, 202.6°E with A_{95}=6.2° (Gilder and Courtillot, 1997). The paleomagnetic pole for Sikhote Alin is west of that for the North China Block (Fig. 4). The angular distance between the poles (15.4°±11.9°) is statistically significant. A relative tectonic movement is claimed between Sikhote Alin and North China Block. The relative movement is described as a counterclockwise rotation through

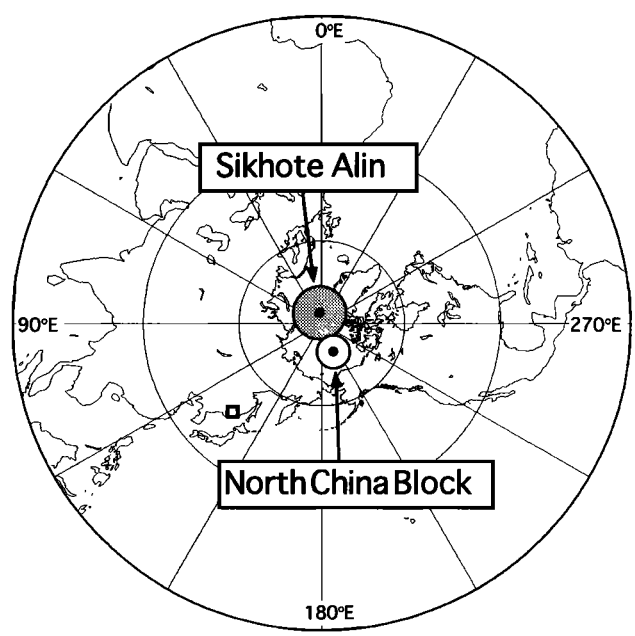


Fig. 4. Cretaceous paleomagnetic poles (with associated 95% confidence circles) for Sikhote Alin and North China Block. The open square denotes the Sikhote Alin reference point.

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20.5°±17.7° with respect to North China Block, according to the calculation method in Otofuji et al. (1998).

The relative movement of Sikhote Alin occurred later than 66 Ma, and ceased by 51 Ma when the Bogopol Group was formed. The pole of the Bogopol Group (78.9°N, 236.7°E with A_{95}=10.3°) (Otofuji et al., 1995) is in good agreement with that of the North China Block (83.8°N, 199.4°E with A_{95}=7.1°) (Gilder and Courtillot, 1997).

The timing of the relative movement of Sikhote Alin agrees with that of the fault motion along the central Sikhote Alin fault. Kinematic analysis of stretching lineation and $^{40}Ar/^{39}Ar$ dating of biotite and muscovite in metamorphic rocks revealed that the sinistral fault motion along the central Sikhote Alin fault occurred between 73 and 58 Ma (Faure et al., 1995). The sinistral motion brought about the counterclockwise rotation of Sikhote Alin.

The counterclockwise rotation is anticipated for blocks demarcated by the central Sikhote Alin fault and its branch faults. At least five blocks are identified within our study area according to the geological map of Bazhanov and Oleinik (1986): four branch faults trend towards WNW from Kuzunetsovo, Malaya Kema, Rudnaya Pristan and Kamen Ribolov respectively (Fig. 1b). The deformed zone with a half width of about 100 km along the central Sikhote Alin fault is expected. This aspect is consistent with that of the San Andreas fault where the deformed zone is also spread out over a half width of 100 km (Luyendyk et al., 1985). A Cretaceous counterclockwise tectonic rotation is also observed in Heilongjiang Province, northeastern China (Uchimura et al., 1996). This rotation is ascribed to sinistral motion along the northeastern extension of the Tan-Lu fault. The tectonic feature in the Northeast Asian continental margin during the Late Cretaceous to Paleogene is characterized by deformation associated with fault activity.

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